

HAILUN, China

WMO: 507560

Lat: 47.4453N

Lon: 126.8744E

Elev: 722

StdP: 14.32

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-23.4	-19.7	-29.2	1.1	-22.5	-25.6	1.4	-19.0	16.2	6.1	14.1	5.0	2.6	160	0.554

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	14.8	86.6	67.9	83.8	68.5	81.3	68.1	74.7	80.7	73.0	78.6	71.5	77.3	9.2	180

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
72.9	125.4	77.7	71.2	118.1	76.4	69.5	111.3	75.0	38.9	80.9	37.3	78.4	35.9	77.5	82.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
18.7	16.1	14.0	-26.7	92.5	6.0	3.4	-31.0	94.9	-34.5	96.9	-37.9	98.8	-42.2	101.3
			-26.9	77.5	5.9	1.8	-31.2	78.7	-34.7	79.8	-38.0	80.8	-42.3	82.0
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	36.9	-5.6	3.8	21.9	41.9	56.6	67.9	72.0	68.2	57.1	40.5	17.6	-1.2	(6)
(7)		DBStd	28.58	8.32	9.74	11.41	8.38	7.60	6.34	4.02	4.60	6.67	8.57	11.89	8.87	(7)
(8)		HDD50	7069	1724	1292	872	269	23	0	0	16	314	973	1586		(8)
(9)		HDD65	10757	2189	1712	1337	694	278	40	2	21	249	761	1423	2051	(9)
(10)		CDD50	2283	0	0	0	25	228	538	682	565	227	18	0	0	(10)
(11)		CDD65	498	0	0	0	0	19	128	220	121	10	0	0	0	(11)
(12)		CDH74	3561	0	0	0	10	258	1138	1402	657	95	1	0	0	(12)
(13)		CDH80	848	0	0	0	0	59	376	321	81	11	0	0	0	(13)
(14)	Wind	WSAvg	6.2	4.7	5.4	6.8	8.0	8.1	6.7	5.9	5.6	6.0	6.7	6.0	5.0	(14)
(15)	Precipitation	PrecAvg	21.7	0.1	0.1	0.3	0.9	1.9	3.8	5.8	5.2	2.6	0.9	0.3	0.2	(15)
(16)		PrecMax	34.1	0.2	0.3	0.9	2.5	4.3	8.5	17.6	10.4	5.2	2.4	1.0	0.7	(16)
(17)		PrecMin	13.7	0.0	0.0	0.0	0.0	0.3	0.8	1.2	0.6	0.2	0.0	0.0	0.0	(17)
(18)		PrecStd	4.6	0.1	0.1	0.2	0.7	1.1	1.8	3.1	2.8	1.3	0.6	0.2	0.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	21.0	35.7	54.0	75.0	86.1	93.3	90.1	84.6	81.2	69.1	49.4	28.7	(19)
(20)		MCWB	18.6	29.6	41.9	53.3	59.4	67.7	72.5	72.5	65.1	54.6	42.3	26.8		(20)
(21)		2%	DB	15.3	29.8	47.5	67.4	80.8	87.8	86.0	82.3	76.5	62.8	43.4	20.6	(21)
(22)		MCWB	13.5	25.5	37.3	49.0	58.0	66.2	71.1	71.7	63.1	51.8	37.4	18.7		(22)
(23)		5%	DB	10.5	24.4	42.5	62.1	76.2	84.3	83.5	80.1	73.3	58.1	38.2	16.2	(23)
(24)		MCWB	8.9	21.1	34.4	46.8	56.5	65.2	71.4	70.1	60.9	48.3	33.1	14.6		(24)
(25)		10%	DB	6.8	19.1	38.2	57.0	71.7	80.7	81.1	77.8	69.9	54.1	33.8	12.4	(25)
(26)		MCWB	5.6	16.4	31.9	43.9	54.4	64.0	70.7	70.7	68.6	59.4	46.1	30.4	10.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	18.6	30.6	44.0	54.8	64.2	72.4	77.4	76.5	69.7	57.4	43.1	26.4	(27)
(28)		MDCB	20.7	34.5	52.1	70.4	78.3	83.1	82.6	80.8	75.2	65.3	47.5	28.2		(28)
(29)		2%	WB	13.5	25.4	38.3	50.9	60.8	70.1	75.3	74.3	65.9	53.2	37.6	19.0	(29)
(30)		MDCB	15.2	29.6	45.3	64.1	73.9	80.8	81.8	79.0	72.5	60.2	42.7	20.6		(30)
(31)		5%	WB	9.0	20.9	35.1	47.7	58.3	68.3	73.8	72.5	63.4	50.0	33.7	14.6	(31)
(32)		MDCB	10.5	24.3	41.4	59.8	71.7	78.2	79.8	77.4	70.1	55.9	37.1	16.1		(32)
(33)		10%	WB	5.6	16.4	32.1	44.9	56.1	66.6	72.4	70.8	61.1	46.8	30.2	10.9	(33)
(34)		MDCB	6.8	19.0	37.9	55.2	69.0	76.0	78.2	75.8	67.6	53.1	33.8	12.3		(34)
(35)	Mean Daily Temperature Range		MDBR	15.6	18.3	17.9	19.3	20.1	17.8	14.8	15.8	20.0	17.7	15.6	14.7	(35)
(36)		5% DB	MCDBR	17.9	20.7	21.6	27.7	27.0	23.5	18.7	18.7	24.3	23.3	19.8	16.9	(36)
(37)		MCWBR	16.5	17.4	14.2	15.0	11.8	8.7	7.9	9.3	11.6	13.8	14.6	15.7	(37)	
(38)		5% WB	MCDBR	17.7	20.5	19.8	24.1	22.8	18.5	14.7	14.9	19.1	20.3	18.6	17.1	(38)
(39)		MCWBR	16.4	17.4	13.5	14.0	11.4	8.3	8.2	8.9	11.0	13.7	14.5	16.0		(39)
(40)	Clear-Sky Solar Irradiance	taub	0.300	0.324	0.387	0.485	0.547	0.566	0.540	0.476	0.431	0.413	0.342	0.306	(40)	
(41)		taud	2.134	2.084	1.991	1.866	1.784	1.837	1.929	2.082	2.122	2.045	2.152	2.125	(41)	
(42)		Ebn at Noon	238	259	258	243	233	229	233	244	243	221	218	217	(42)	
(43)		Edh at Noon	28	36	47	59	67	64	57	47	41	38	28	25	(43)	
(44)	All-Sky Solar Radiation	RadAvg	620	994	1380	1545	1736	1797	1661	1529	1292	882	576	475	(44)	
(45)		RadStd	42	58	101	140	155	218	168	115	113	64	47	31	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days				
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65	
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(46)
		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (4 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page